

LESSON 6-8 • SESSION 2 • PRACTICE • FORM B

Expanding and Factoring Expressions

Learning target – I can expand an expression and factor one using the greatest common factor. · 6.EE.A.3

REMEMBER

- To **expand**, multiply the outside number by each term inside: $3(2x + 5) = 6x + 15$.
- To **factor**, pull out a number every term shares. Factoring **undoes** expanding.
- Use the **GCF**: $8x + 20 = 2(4x + 10)$ is not finished, because 4 and 10 still share 2. $8x + 20 = 4(2x + 5)$ is.

WORKED EXAMPLEFactor $18x + 27$ using the GCF.**Step 1** The GCF of 18 and 27 is 9.**Step 2** Divide each term: $18x \div 9 = 2x$ and $27 \div 9 = 3$.**Step 3** Expand to check: $9 \times 2x + 9 \times 3 = 18x + 27$.**Answer** $9(2x + 3)$ **WORDS** **expand** (*desarrollar*) · **factor an expression** (*factorizar una expresión*) · **greatest common factor** (*máximo común divisor*)**PRACTICE**

show your work · label units

- 1.**
- Expand each expression.

a. $9(w + 4) =$ _____

b. $3(4d + 5) =$ _____

c. $5(y + 9) =$ _____

d. $8(n + 3) =$ _____

- 2.**
- Factor each expression using the GCF.

a. $24q + 18 =$ _____

b. $21k + 28 =$ _____

c. $10b + 25 =$ _____

d. $12x + 30 =$ _____

- 3. Art Club Costs.**
- Club 1 writes its supply cost as
- $6(s + 8)$
- . Club 2 writes its cost as
- $28s + 49$
- .

Hint Find the GCF of 28 and 49 first.

a. Expand Club 1's cost. _____

b. GCF of $28s$ and 49 : _____

c. Factor Club 2's cost. _____

Show your work.

- 4. Expand, Then Combine.**
- Simplify
- $2(x + 9) + 4x$
- .

Hint 2 times 9 is part of the answer.

Show your work.

Simplest form _____

- 5. Expand, Then Combine.**
- Simplify
- $5(z + 3) + 3z$
- .

Show your work.

Simplest form _____

6. **Omar's Factoring.** Omar factored $18y + 45$ as $3(6y + 15)$. Is he done? If not, factor with the GCF.

Hint Do 6 and 15 still share a factor?

Show your work.

GCF form _____

7. **Both Directions.** Factor $9a + 27$ using the GCF. Then expand your answer to show the two forms match.

Hint 9 divides both 9 and 27.

Show your work.

Factored form _____

8. **Garden Rows.** A garden has $18p + 30$ plants.

- Factor the expression using the GCF. _____
- What could the factored form tell you about rows? _____
- Find the number of plants when $p = 4$. Use both forms. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Amina's Work.** Amina expanded $7(k + 3)$ and wrote $7k + 3$. She said the 3 was already inside the parentheses. Explain her mistake. Then expand correctly.

TEST PRACTICE

choose the best answer

10. Which is $8n + 12$ factored using the GCF?

A $2(4n + 12)$ **B** $4(2n + 3)$ **C** $8(n + 4)$ **D** $4(2n + 12)$

11. To factor an expression with the GCF, you pull out the –

A greatest common factor **B** least common multiple
C largest term **D** variable

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

6x + 9 Q	7 O	6x + 18 D	12d + 15 P	9(2y + 5) C	8z + 15 U	8z + 3 Z	6(4q + 3) R	102 T
---------------------------	----------------------	----------------------------	-----------------------------	------------------------------	----------------------------	---------------------------	------------------------------	------------------------

#1b	#2a	#3b	#4	#5	#6	#8c
-----	-----	-----	----	----	----	-----

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____